

■ Observers

2b(0-3)

a — attention bucket descriptor metadata

1b

b_r — **recency**
line-local hit/promote

1b

b_q — **structured-query**
descriptor class

1b

b_s — **service-critical**
desc. class $\geq$ fixed thr.

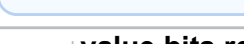
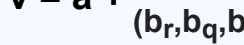
1b

b_u — **reuse**
address-reuse state

■ Bit-lattice Fusion

2-bit

3-bit add



$v = a + \text{popcount}(b_r, b_q, b_s, b_u)$

no MAC · no CAM

■ Contextual Floor

3b, reg.



fixed T_f

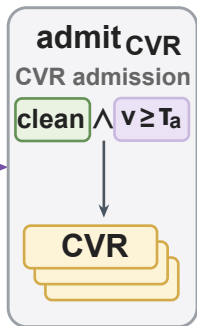
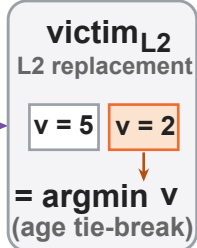
risk signature
small capacity

late phase

struct-query
dominant

drives

drives



→ value / data path
- - - replacement / admission

value bits registered one stage early ; read on fill/
eviction / CVR admission—never on the L1-hit path